

PARUL UNIVERSITY
FACULTY OF ENGINEERING & TECHNOLOGY
B.Tech. Summer 2018 - 19 Examination

Semester: 5
Subject Code: 03105303
Subject Name: Theory of Computation

Date: 18/05/2019
Time: 10:30 am to 01:00pm
Total Marks: 60

Instructions:

1. All questions are compulsory.
2. Figures to the right indicate full marks.
3. Make suitable assumptions wherever necessary.
4. Start new question on new page.

Q.1 Objective Type Questions - (Part 1 to 5 is MCQ, Part 6 to 10 is Fill in the blanks and Part 11 to 15 is one word answer (All are compulsory) (Each of one mark) **(15)**

1. Context-free languages are closed under:
 - A) Union, intersection
 - B) Union, Kleene closure
 - C) Intersection, complement
 - D) Complement, Kleene closure
2. Minimum length of the string formed by regular expression $(a+b)^* aba^* (a+b)^*$
 - A) 2
 - B) 6
 - C) 4
 - D) 8
3. A regular grammar is
 - A) Type-0
 - B) Type-1
 - C) Type-2
 - D) Type-3
4. Regular expression for all strings starts with ab and ends with bba is.
 - A) aba^*b^*bba
 - B) $ab(ab)^*bba$
 - C) $ab(a+b)^*bba$
 - D) All of the mentioned
5. In the context-free grammar below, S is the start symbol, a and b are terminals, and ϵ denotes the empty string.
 $S \rightarrow aSa \mid bSb \mid a \mid b \mid \epsilon$
 Which of the following strings is NOT generated by the grammar?
 - A) aaaa
 - B) baba
 - C) abba
 - D) babaaabab
6. There are _____ tuples in DFA.
7. Transition function of NDFSA maps $\delta =$ _____
8. Grammar that produce more than one derivation tree for same sentence is _____
9. The language accepted by a Push down Automata is _____ language.
10. According to Arden's theorem if every regular expression is in the form of $R = Q + RP$ then there is a unique solution is _____
11. The output of the Mealy machine is determined only by its current state. It's True or False?
12. Write the condition for the left recursive grammar.
13. If $L = \{a^n b^n \mid n > 0\}$ it is not a Regular Language. True or False?
14. Any regular language has an equivalent CFG. It's True or False?
15. All languages can be generated by context-free grammar. True or False?

Q.2 Answer the following questions. (Attempt any three) **(15)**

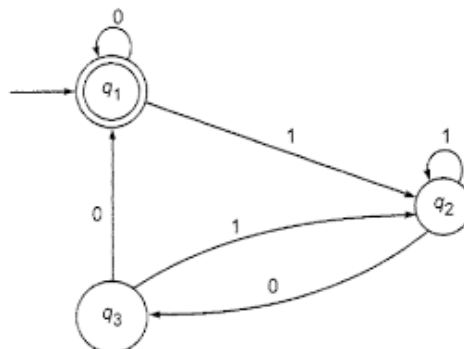
A) Convert the following CFG to CNF :

$S \rightarrow ABA$

$A \rightarrow aA / \epsilon$

$B \rightarrow bB / \epsilon$

B) Define ARDEN's Theorem. Construct a regular expression corresponding to the automata given below using ARDEN's Theorem.



C) Convert the following Mealy machine into its equivalent Moore machine:

Present State	Next State			
	Input 0		Input 1	
	State	Output	State	Output
q ₀	q ₁	N	q ₂	N
q ₁	q ₁	Y	q ₂	N
q ₂	q ₁	N	q ₂	Y

D) Using principle of Mathematical induction, Prove that for every $n \geq 1$,

$$1 + 3 + 5 + \dots + (2n - 1) = n^2$$

Q.3 A) Define ambiguous grammar. Also Prove that the following language is ambiguous and convert into unambiguous: (07)

$$X \rightarrow X + X \mid X * X \mid a$$

B) Design a Turing Machine for the following Language: (08)

$$L = \{a^n b^n \mid n \geq 0\}$$

OR

B) Convert the grammar into Greibach Normal Form: (08)

$$S \rightarrow AB$$

$$A \rightarrow BS / b$$

$$B \rightarrow SA / a$$

Q.4 A) Construct a PDA for the following Language: (07)

$$L = \{a^n b^m c^n \mid m, n \geq 0\}$$

OR

A) 1) For the following Regular Expression draw a DFA recognizing the corresponding language. (07)

$$(0 + 1)^* 10 (0+1)^*$$

4 Marks

2) Explain Multi-Tape Turing Machine.

3 Marks

B) Define language. Draw Deterministic Finite Automata for the following languages (08)

i) $L_1 = \{x \in (0,1)^* \mid x \text{ contains } 101 \text{ as a substring}\}$

ii) $L_2 = \{x \in (0,1)^* \mid x \text{ contains odd number of zero}\}$

iii) $L_3 = \{x \in (0,1)^* \mid x \text{ ends with } 11\}$

iv) $L_4 = \{x \in (0,1)^* \mid x \text{ start with } 001\}$